

THESIS__GUIDELINES.md

Bachelor Thesis Guidelines — FOM Wirtschaftsinformatik

Version: Updated based on FOM Guidelines (2024), Prof. Dr. Dieter Litzinger Notes, Methodology References, and Current Exposé.

1. General Scientific Principles

Scientific work at FOM must:

- be evidence-based,
- follow a systematic and fact-oriented methodology,
- use appropriate scientific research methods,
- derive methodology from the research question,
- ensure comprehensibility and transferability of results,
- maintain objectivity and neutral language.

The work must demonstrate that the student can independently solve a problem using scientific methods within a limited timeframe.

Key evaluation criteria:

- Scientific rigor
- Methodological appropriateness
- Logical structure
- Neutral argumentation
- Literature quality
- Correct citation style
- Formal correctness
- Practical relevance

Sources: - FOM Leitfaden 2024 - Vorbereitungsseminar Bachelor Thesis (Prof. Dr. Dieter Litzinger) - Kurzgutachten Bewertungskriterien

2. Formal Structure of the Thesis

The thesis should contain the following elements in this order:

1. Title Page
2. Blocking Notice (if applicable)
3. Table of Contents
4. List of Figures (if applicable)

5. List of Tables (if applicable)
 6. List of Abbreviations (if applicable)
 7. Main Text
 8. Bibliography / References
 9. AI Usage Documentation (KI-Verzeichnis)
 10. Appendices (if applicable)
 11. Declaration of Academic Integrity
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3. Required Thesis Structure

3.1 Introduction

The introduction must contain:

Introduction / Context

- Motivation
- Context of the topic
- Research gap
- Practical relevance

Problem Statement

- Clearly defined research problem
- Research question(s)
- Explanation of why the problem is relevant

Objective of the Thesis

- Clear objective
- Scope and limitations
- Expected contribution

Methodological Approach

The methodology must explicitly explain:

- what method is used,
- why this method was selected,
- how the method will be applied.

The procedural flow of the thesis must also be described in prose.

3.2 Foundations / Theoretical Background

This section should:

- define important concepts,
- explain technological and theoretical foundations,
- provide scientific background literature,
- establish the conceptual basis for later analysis.

For the current thesis topic this includes:

- Artificial Intelligence
 - Machine Learning
 - Digital Information Systems
 - Monitoring Systems
 - Algorithmic Governance
 - Platform Economies
 - Recommender Systems
 - Data-driven Decision Systems
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3.3 Main Analysis / Investigation

The main chapters must:

- answer the research question,
- systematically analyze the topic,
- apply scientific literature critically,
- connect technological foundations with socioeconomic implications.

The argumentation must remain:

- neutral,
 - evidence-based,
 - logically structured,
 - scientifically grounded.
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3.4 Conclusion

The conclusion must include:

Summary of Findings

- concise synthesis of results,
- answer to the research question,
- interpretation of findings.

Critical Reflection

- limitations of the work,
- methodological limitations,
- possible biases in literature.

Outlook / Future Research

Only include if new research questions emerge.

4. Language and Style Requirements

The thesis must:

- use neutral scientific language,
- avoid emotional or political wording,
- avoid colloquial expressions,
- avoid first-person formulations (“I”, “we”),
- avoid rhetorical questions,
- maintain grammatical correctness.

Quotes must use language-appropriate quotation marks:

- German: „...“
- English: “...”

Quotation marks must only be used for actual quotations.

5. Methodological Requirements

5.1 Current Thesis Methodology

The current thesis follows:

- a literature-based,
- conceptual-analytical,
- non-empirical research approach.

No empirical data collection or software development is performed.

The work focuses on:

1. Analysis of technological foundations
2. Identification of structural system characteristics
3. Analysis of socioeconomic implications
4. Systematic categorization and synthesis

5.2 Literature Review Requirements

The literature review must:

- be systematic,
- document databases used,
- document keywords used,
- document selection criteria,
- describe inclusion/exclusion logic,
- allow replicability.

The literature search process itself is part of scientific rigor.

The following must be explicitly documented:

- databases,
- keywords,
- publication periods,
- filtering logic,
- backward/forward search procedures.

Important references:

- vom Brocke et al.
 - Webster & Watson
 - Levy & Ellis
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5.3 Qualitative Research Considerations

Even though the thesis is literature-based, the following principles apply:

- systematic categorization,
- transparent analysis procedure,
- intersubjective traceability,
- structured interpretation.

Thematic categorization should follow clear analytical rules.

6. Research Questions and Hypotheses

Central Research Question

How do AI-supported monitoring systems function as digital information systems and which socioeconomic implications are discussed in scientific literature?

Hypotheses

H1

AI-supported monitoring systems are characterized by multi-layered architectures consisting of:

- data collection,
- analysis,
- evaluation,
- algorithmic selection.

H2

Machine learning integration creates data-driven decision and evaluation structures that supplement or partially replace traditional control mechanisms.

H3

Scientific literature attributes significant socioeconomic implications to AI-supported monitoring systems, especially regarding:

- information selection,
 - behavioral influence,
 - governance structures.
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7. Literature Research Requirements

Databases

The literature search currently includes:

- Google Scholar
 - SpringerLink
 - IEEE Xplore
 - ACM Digital Library
 - Web of Science
 - ScienceDirect
 - Wiley Online Library
 - Emerald Insight
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Selection Criteria

Sources should be selected based on:

- peer review,
 - scientific quality,
 - topical relevance,
 - recency,
 - interdisciplinary relevance.
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Recommended Source Types

Preferred:

- Peer-reviewed journal articles
- Scientific conference papers
- Academic books
- Springer publications
- IEEE papers
- AIS literature

Use cautiously:

- Business articles
- Non-reviewed web pages
- Media commentary

Avoid:

- unsourced blogs,
 - opinion pieces without scientific basis,
 - low-quality AI-generated summaries.
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8. Structural Recommendations for Current Thesis

Recommended Final Chapter Structure

1. Introduction
 2. Technological and Theoretical Foundations
 3. AI-supported Monitoring Systems in Digital Information Environments
 4. Socioeconomic Implications
 5. Discussion
 6. Conclusion and Outlook
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9. Formatting and Styling Guidelines

9.1 General Formatting

The formatting of the thesis must remain consistent throughout the entire document.

Recommended formatting based on FOM guidelines:

- Font: Times New Roman or Arial
- Font size:
 - Body text: 11 pt or 12 pt
 - Footnotes: 10 pt
- Line spacing:
 - 1.5 lines in body text
 - single-line spacing in footnotes
- Text alignment: justified
- Margins:
 - Left: 3–4 cm
 - Right: 2–2.5 cm
 - Top: 2–2.5 cm
 - Bottom: 2–2.5 cm
- Automatic page numbering required
- Consistent heading hierarchy required

Recommended paragraph formatting:

- No excessive spacing between paragraphs
- Consistent spacing before/after headings
- New sections should begin clearly separated

9.2 Scientific Writing Style

The thesis should:

- prioritize precision over rhetorical style,
- formulate arguments analytically,
- avoid emotional wording,
- avoid ideological framing,
- distinguish clearly between facts, interpretation, and literature positions.

Preferred style:

- concise sentences,
- technically precise terminology,
- evidence-supported claims,
- structured transitions between sections.

Avoid:

- overly dramatic wording,
 - activist language,
 - unsupported assumptions,
 - broad generalizations.
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9.3 Page Structure

Each chapter should:

- begin with a short introduction,
- maintain logical flow,
- end with a short transition or interim conclusion when appropriate.

Recommended structure within analytical chapters:

1. Introduction of concept
 2. Literature overview
 3. Critical analysis
 4. Comparison or synthesis
 5. Transition to next section
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9.4 Citations and Footnotes

Footnotes should:

- remain concise,
- contain supporting references,
- avoid extensive side discussions.

Direct quotations:

- only when necessary,
- must include exact page numbers,
- should not dominate the thesis.

Paraphrasing is generally preferred over excessive direct quotation.

9.5 Tables and Figures Formatting

Tables and figures must:

- use consistent formatting,
- maintain readable font sizes,
- include captions,
- include source references,

- be referenced in text before appearance.

Captions:

- Tables → caption above table
- Figures → caption below figure

Example:

- Figure 3: Multi-layered Monitoring Architecture
 - Source: Own illustration based on Kitchin (2017)
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9.6 Bibliography Formatting

The bibliography must:

- use one consistent citation style,
- maintain alphabetical order,
- include complete bibliographic information,
- contain DOI links where possible.

Missing metadata is considered a formal weakness.

9.7 AI Usage Documentation

Current FOM guidelines require documentation of AI-generated content.

The thesis should therefore include:

- a KI-Verzeichnis,
- documentation of significant AI-supported work,
- potentially selected prompts in the appendix if required by the supervisor.

AI tools may support:

- language improvement,
- structuring,
- brainstorming,
- summarization.

However, scientific responsibility remains fully with the student.

9.8 Headings

Headings

- Decimal numbering required

- Example:
 - 1
 - 1.1
 - 1.1.1
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Figures

Figures must:

- be numbered continuously,
- contain captions below the figure,
- be referenced in text before usage,
- include proper citations.

Example:

“Figure 3 shows the structural interaction between platform architecture and behavioral analysis.”

Tables

Tables must:

- be numbered continuously,
 - contain captions,
 - be referenced in text,
 - maintain consistent formatting.
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Citations

The thesis may use:

- Footnote citation style,
- Bracket citation style,
- Short-reference citation style.

The citation style must remain consistent throughout the thesis.

All AI-generated content must be documented according to current FOM requirements.

10. Common Mistakes to Avoid

Methodology Problems

- Missing methodological justification
- Vague research questions
- Lack of analytical structure
- Pure description without analysis

Literature Problems

- Weak source quality
- Too many internet sources
- Missing peer-reviewed literature
- No documented search strategy

Language Problems

- Emotional wording
- Political bias
- Informal language
- “I think” formulations

Structural Problems

- No clear chapter logic
 - Missing transition between sections
 - Weak conclusion
 - No critical reflection
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11. Evaluation Criteria Used by FOM Reviewers

The thesis will likely be evaluated according to:

Formal Criteria

- completeness,
- formatting,
- spelling and grammar,
- citation consistency,
- bibliography correctness.

Methodological Criteria

- methodological appropriateness,

- methodological transparency,
- interpretive quality,
- reproducibility.

Content Criteria

- logical structure,
 - problem relevance,
 - argumentation quality,
 - literature quality,
 - scientific depth,
 - independent analytical contribution.
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12. Current Thesis Positioning

The current thesis is positioned within:

- Wirtschaftsinformatik,
- digitale Informationssysteme,
- algorithmic governance,
- AI-supported decision systems,
- platform economies,
- data-driven monitoring architectures.

The work combines:

- technological system analysis,
 - information systems theory,
 - socioeconomic implications.
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13. Important Scientific References for the Thesis

Design Science / IS Research

- Hevner et al. (2004)
- Wilde & Hess (2006)
- Österle et al. (2010)

Literature Review Methodology

- vom Brocke et al.
- Webster & Watson
- Levy & Ellis

Qualitative Analysis

- Mayring
- Mayring & Fenzl

AI / Monitoring / Governance

- Kitchen
- Yeung
- Russell
- Zuboff

14. Key Dates

Task	Start	End
Besprechung Exposé	01.03.	01.03.
Anmeldung Thesis	05.03.	05.03.
Erweiterte Literaturrecherche	01.03.	14.03.
Schreiben Kapitel 2 (Grundlagen)	15.03.	28.03.
Schreiben Kapitel 3 (Systemanalyse)	29.03.	15.04.
Schreiben Kapitel 4 (Implikationen)	16.04.	30.04.
Diskussion & Fazit	01.05.	10.05.
Korrekturphase	11.05.	25.05.
Druck & Abgabe	26.05.	30.05.2026

15. Final Recommendations

The thesis should:

- remain analytically focused,
- avoid ideological positioning,
- maintain strong scientific rigor,
- document the literature review transparently,
- consistently connect technical system structures with socioeconomic implications,
- emphasize Wirtschaftsinformatik relevance,
- demonstrate independent analytical synthesis.

The strongest version of the thesis will not merely describe surveillance or AI systems, but systematically analyze how digital information systems structure decision-making, behavioral steering, and governance mechanisms within platform-based environments.